

Adaptive Human-LLMs Interaction Collaboration: Reinforcement Learning driven Vision-Language Models for Medical Report Generation

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Abstract

The large language models (LLMs) and vision models have demonstrated great potential in the field of healthcare. The integration of vision-language models (VLMs) has led to significant advancements in automatic medical report generation. However, effectively incorporating doctor feedback into the report generation process remains an unresolved challenge in clinical practice. In this paper, we propose an adaptive human-LLMs collaborative vision-language model that dynamically adjusts the report generation process based on the feedback of doctors and LLMs. We designed a reinforcement learning module to dynamically update the generated content generated by the VLMs according to doctor feedback, addressing the challenge of integrating doctor feedback into the generation model. To mitigate over-reliance on doctor feedback, we introduced the RL-LLM module, which uses an LLM to assess the quality of the generated report. The model adopts a participatory design approach in collaboration with clinical doctors to improve report accuracy and clinical availability.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Health informatics**.

Keywords

Human LLMs interaction, Medical report generation, Health informatics

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1 Introduction

The increasing availability of medical imaging technologies has significantly improved diagnostic capabilities in healthcare. However, the process of generating medical reports from these images remains a labor-intensive and error-prone task. Despite advancements in automated systems for medical image analysis, automated medical report generation is still fraught with challenges, including incomplete or inaccurate descriptions of key lesions, anomalies, or other critical findings in medical images. The reliance on human expertise for creating high-quality reports is indispensable but limits scalability and efficiency. Thus, it is crucial to explore methods that combine the strengths of both artificial intelligence (AI) and human expertise to optimize medical report generation.

This paper addresses two key research questions (RQs) essential for improving medical report generation through AI collaboration. The first research question asks, How can professional doctors' feedback be effectively integrated into the report generation process to enhance report quality? Feedback from medical professionals is crucial for improving the accuracy and completeness of generated reports. Doctors possess domain expertise that is indispensable for verifying the content generated by AI models, particularly when it comes to critical and nuanced aspects of medical reports. However, the challenge lies in efficiently incorporating this feedback into the model without introducing excessive manual intervention or reducing the automation's effectiveness. The second research question, How can we ensure that the model dynamically adjusts the generated content based on doctors' feedback without over-relying on the doctor and thus reducing the degree of automation? tackles the balance between automation and human intervention. While doctors' input is necessary for refining AI-generated reports,

over-reliance on manual corrections can negate the benefits of automation. It is crucial to design systems that learn from feedback to improve over time, reducing the need for continuous human input.

To address these challenges, we propose an adaptive human-LLMs collaborative vision-language model that integrates doctors' feedback into the report generation process and progressively optimizes its output. The key goal of this model is to create an interactive, clinician-in-the-loop tool that not only automates the report generation process but also learns from clinician feedback to continuously improve its outputs. The model consists of three main modules: the generation module, which employs a multimodal vision-language model to generate initial drafts of medical reports from images, the reinforcement learning module, which learns from doctors' modifications and adjusts the report generation process accordingly, and the large language model distillation module, which uses LLMs to evaluate and refine the generated reports automatically, minimizing dependency on human input. Specifically, the generation module employs a vision-language model that encodes medical images and generates textual descriptions through a Transformer-based architecture. The model integrates both image features and textual context through attention mechanisms, allowing it to focus on relevant areas of the image for more accurate descriptions. This approach, commonly used in image captioning, enables the model to generate medically relevant textual content directly from medical images, which can then be iteratively refined. The reinforcement learning module addresses the issue of dynamic model adjustment. By incorporating doctor feedback as a reward signal, the system continuously improves its ability to generate high-quality reports. Doctors provide corrections or suggestions, and the model optimizes its reporting strategies based on these interactions. The reward model is trained to associate positive feedback with successful report modifications, guiding the system toward better performance. The large language model distillation module (RL-LLM) represents a novel approach to reducing human intervention. Instead of manually annotating the feedback or training reward models, the system uses a pre-trained LLM to evaluate the quality of generated reports and provide reward signals. This approach streamlines the process, making it scalable and more efficient by leveraging the powerful capabilities of large language models in assessing text quality.

This paper outlines the design and implementation of these modules, discussing their integration within an adaptive system that learns from feedback while maintaining high levels of automation. The adaptive human-LLMs collaborative framework can improve both the accuracy of medical reports and the efficiency of the report generation process, ultimately benefiting both healthcare providers and patients.

2 Related Work

2.1 Multimodal Large Models in Healthcare

The integration of multimodal large models in healthcare is advancing the way AI supports medical practice by combining diverse data types, such as clinical text, medical images, and sensor data, to improve decision-making. However, deploying these models in clinical environments comes with challenges related to biases, regulatory compliance, and trust in AI outcomes.

Foundation models, such as GPT-3 and GPT-4, have been identified as transformative in healthcare, consolidating knowledge across disciplines and performing a wide range of tasks from clinical question answering to report summarization. However, these models pose risks related to transparency, bias, and the accuracy of their outputs in healthcare contexts [2], [18]. Despite these challenges, LLMs are showing promise in medical question answering, achieving near-expert-level performance [19], [20].

Multimodal models in medical imaging are particularly useful for automating report generation. For instance, Hyland et al. introduced MAIRA-1, which combines medical images and text for generating context-aware radiology reports [9]. Leclerc et al. further demonstrated the potential of GPT-based models to improve diagnostic accuracy by merging image recognition and NLP [12]. Xu et al. proposed ELIXR, a system combining LLMs with radiology vision encoders for X-ray analysis, expanding the use of LLMs in medical imaging [21].

In biomedical knowledge extraction, Gu et al. illustrated how LLMs can distill knowledge from medical literature to identify risks, such as adverse drug events [8], and Preston et al. showed how deep learning can structure oncology data to extract valuable patient-specific insights [16]. While these advances are promising, regulatory and trust issues remain pivotal for the effective integration of multimodal models in healthcare. Gilbert et al. suggested that LLMs must be treated as medical devices and undergo rigorous regulatory review [6]. These regulatory concerns are critical to ensuring the safety and efficacy of AI in healthcare.

The interpretability of AI models remains a significant challenge. Nori et al. highlighted that despite the success of models like GPT-4, issues like biases in training data and transparency in decision-making need to be addressed [13]. Furthermore, the need for models to explain their reasoning processes is crucial to ensuring clinician trust, especially in high-stakes medical scenarios [14].

Despite these challenges, multimodal models have shown potential in simplifying medical information for both clinicians and patients. Jeblick et al. demonstrated how ChatGPT can be used to simplify radiology reports, making them more patient-friendly without losing diagnostic accuracy [10]. Krishna et al. developed techniques to automate the generation of SOAP notes, which helps reduce the administrative burden on clinicians [11].

In summary, while multimodal large models show significant promise in healthcare, overcoming challenges such as transparency, bias, and regulatory compliance is crucial for their effective integration into clinical settings.

2.2 Human-AI Collaboration in Healthcare

Human-AI collaboration has emerged as a key area in healthcare, aiming to augment clinicians' decision-making and improve patient outcomes. Successful integration of AI requires designing systems that align with clinicians' needs, are intuitive to use, and maintain transparency in their operations.

User-centered design is essential for the adoption of AI in healthcare. Cai et al. emphasized that AI systems must cater to medical professionals by providing user-friendly interfaces and clear communication, thus ensuring trust and effective use in high-pressure environments [4], [3]. Their work highlights the importance of

designing AI tools that help clinicians manage the uncertainties and imperfections of AI, particularly in complex medical decision-making scenarios.

AI's role in addressing cognitive biases in medical decision-making is another key area of research. Bach et al. showed that AI could help reduce anchoring bias in ophthalmology by presenting alternative diagnoses or highlighting overlooked information [1]. This concept is applicable across various medical domains, suggesting that AI can improve diagnostic accuracy and reduce errors in patient care.

Ensuring fairness and transparency in AI systems is also critical. Chien et al. discussed the importance of multi-disciplinary fairness considerations in clinical trials to mitigate biases in data and algorithms [5]. AI systems must be designed to avoid reinforcing existing disparities, ensuring equity and accountability.

Successful integration of AI also requires understanding the sociotechnical dynamics of healthcare. Procter et al. and Zajac et al. emphasized the need for AI systems to be trustworthy and adaptable to the diverse and complex healthcare environments [17], [22]. AI systems must integrate seamlessly into clinicians' workflows, ensuring that they enhance rather than disrupt existing practices. Gu et al. presented XPath, a system designed to support pathologists, which highlights the importance of workflow integration in AI adoption [7].

Regulatory and ethical challenges are paramount in AI deployment. Petersen et al. addressed the need for AI systems to comply with healthcare regulations to ensure patient safety, privacy, and adherence to medical standards [15].

In conclusion, successful human-AI collaboration in healthcare requires that AI systems are designed with clinicians in mind, promote transparency, and are seamlessly integrated into clinical workflows. By addressing issues such as cognitive biases, fairness, and regulatory compliance, AI can effectively augment clinicians' decision-making and improve patient care outcomes. As AI continues to evolve, it will play an increasingly critical role in transforming healthcare delivery.

3 Method

Automated medical image report generation is inherently uncertain and prone to overlooking the description of certain key lesions or areas. We propose two research questions: **RQ1**: How can professional doctors' feedback be effectively integrated into the report generation process to improve report quality? **RQ2**: How can we ensure that the model dynamically adjusts the generated content based on doctors' feedback, avoiding over-reliance on the doctor and thus reducing the degree of automation?

In this section, to address the above issues, we will introduce an adaptive human-AI collaborative vision-language model, which can adjust the report content based on doctors' feedback and progressively optimize its generation process through this feedback.

3.1 Framework Overview

The adaptive human-LLMs collaborative vision-language model, shown in Figure 1, consists of three main modules that work together to enhance the model's performance in report generation

quality. First, the generation module utilizes a multimodal vision-language model to generate draft reports. Second, we design a reinforcement learning module that optimizes the generation process and model parameters based on the content modified by the doctor and their feedback. Third, we develop a large language distillation module that leverages a large language model to learn from the doctor's feedback and adaptively adjust the generated content. Next, we will provide a detailed description of these three key modules.

3.2 Generation Module

The generation module in our system leverages a vision-language model for generating textual medical reports based on given medical images. The architecture follows a Transformer-based design, which is a widely adopted approach for image captioning and text generation tasks. This module consists of three main components: an image encoder, an interaction component, and a language decoder.

The first step involves encoding the input medical image to extract meaningful features. The image encoder is typically implemented by Convolutional Neural Networks (CNNs) or Vision Transformers (ViTs), both of which are effective at processing image data. The CNN or ViT architecture is used to capture the hierarchical features of the medical image. Denote the input medical image as I , and the output of the image encoder as F_{ie} , which represents the feature map of the image:

$$F_{ie} = \text{Encoder}(I), \quad (1)$$

In the case of CNN, this encoding process involves multiple convolutional layers that progressively extract features at various levels of abstraction. For ViT, the input image is divided into fixed-size patches, which are then linearly embedded and processed through a series of transformer layers.

The interaction component is designed to align and integrate both the image features and textual features, allowing the model to focus on specific regions of the image that are most relevant for generating text. The key idea is to use attention mechanisms to establish relationships between image regions and the corresponding textual descriptions. In our scenario, the multi-head attention mechanism enables the decoder to focus on the regions in the image that are relevant to generating the next word in the report, based on the current context of the partially generated text. This interaction allows the model to learn the spatial relationships between image regions and their corresponding textual descriptions.

The language decoder is responsible for generating the final textual output of the medical image report. The decoder follows the standard Transformer architecture and takes the fused features from the image encoder and attention module as input. It generates text in an autoregressive manner, predicting the next word given the current input (which includes both the image features and the previously generated words). Formally, the decoding process is modeled as:

$$p(w_t | w_{1:t-1}, F_{ie}) = \text{Decoder}(w_{1:t-1}, F_{ie}) \quad (2)$$

where w_t denotes the word predicted at time step t , $w_{1:t-1}$ is the previously generated text.

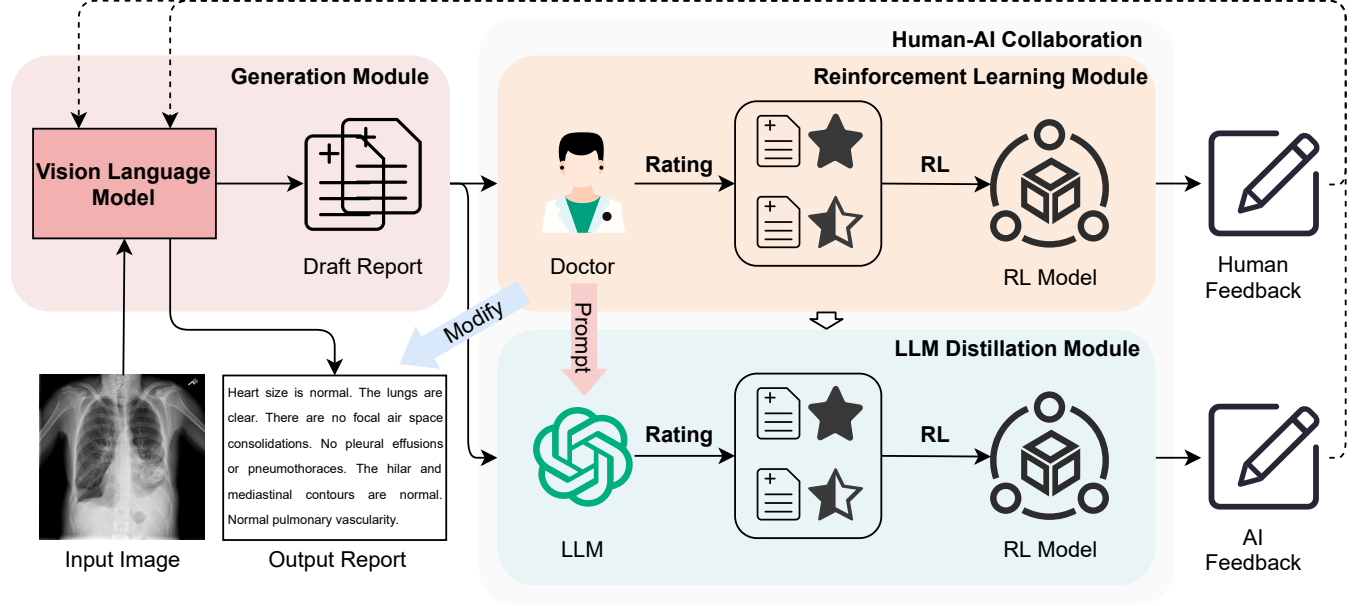


Figure 1: The illustration of the adaptive human-LLMs collaborative multimodal vision-language model

3.3 Reinforcement Learning Module

In the reinforcement learning module, medical professionals first optimize the draft report generated by the generation module with manual feedback. Then, reinforcement learning is applied to learn from the doctor's modifications, optimizing the report generation process. In this setup, the state space consists of the doctor's modifications or selection actions, which can be regarded as the system's states. Each adjustment or selection represents a state, describing the difference between the already generated content in the report and the doctor's intentions. The action space represents the model's "actions," which are adjustments made based on the doctor's modifications or the generation of new report content. The reward signal is based on the doctor's feedback: if the doctor accepts the modification and considers the content accurate, the system receives a positive reward. Conversely, if the report does not meet the doctor's needs or contains errors, a negative reward is given. Finally, the policy gradient method from reinforcement learning is used for policy optimization, ensuring that the model learns the optimal generation strategy after each round of interaction based on the doctor's feedback. A feasible scenario is as follows:

Given an image input i_x , the generation module outputs two draft reports, i.e., r_1 and r_2 . The doctor manually processes the input and draft reports, scoring them according to standards of accuracy, coherence, and rationality in medical reporting. The feedback is annotated in the form of a triple (i_x, r_b, r_w) , where r_b and r_w represent the more preferred and less preferred responses, respectively. The reward model is trained by minimizing the loss function $\mathcal{L}$.

$$\mathcal{L}(\phi) = -\mathbb{E}_{(i_x, r_b, r_w)} \log \sigma \left(r_\phi(i_x, r_b) - r_\phi(i_x, r_w) \right) \quad (3)$$

where σ denotes the sigmoid function. Subsequently, optimization is performed through reinforcement learning to maximize the reward

given by the reward model, which serves as a surrogate for human preferences.

3.4 Large Vision Language Model Distillation Module

To address the challenge of reduced model automation resulting from excessive reliance on doctors, we propose a novel approach that integrates reinforcement learning (RL) with large language model (LLM) feedback, termed RL-LLM. This method leverages the power of large language models to assess the quality of generated draft reports and uses the LLM's evaluation as the reward signal in the reinforcement learning process, thus eliminating the need for labor-intensive human preference annotations and training reward models.

In traditional reinforcement learning, the agent receives a reward signal based on its actions, which it uses to adjust its policy and improve future performance. In our proposed RL-LLM framework, instead of relying on explicit human feedback or hand-crafted reward models, we use the feedback provided by a pre-trained large language model as the reward signal. This allows for the reinforcement learning process to be initiated directly from the evaluation of the generated content by the LLM, which significantly reduces the time and resources typically required for training.

Let s_t represent the state at time t , which corresponds to the draft report generated by the system. The action a_t is the modification or adjustment made to the report at each step. The transition from state to the next state is governed by the action a_t , and the reward is derived from the LLM's evaluation of the report. The reward signal at each timestep is calculated by passing the generated report through the LLM. The LLM produces a quality score, which reflects

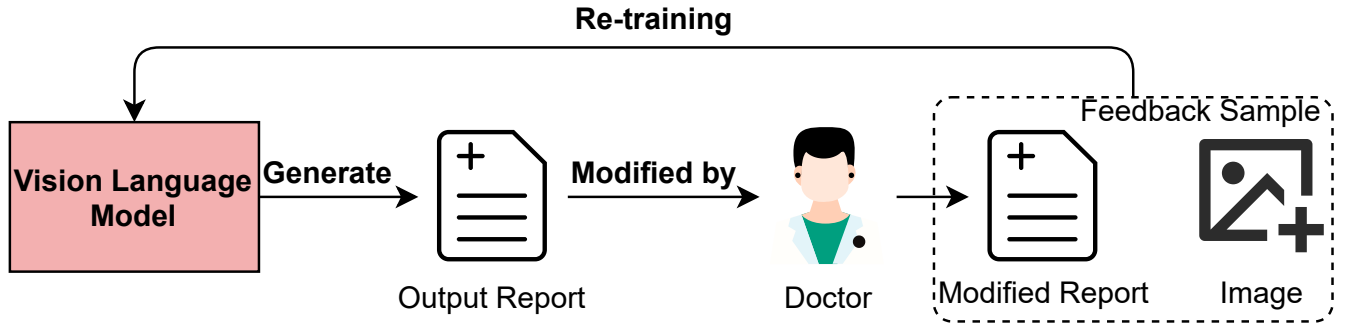


Figure 2: The feedback illustration of doctors on report generation quality in the doctor experience assessment

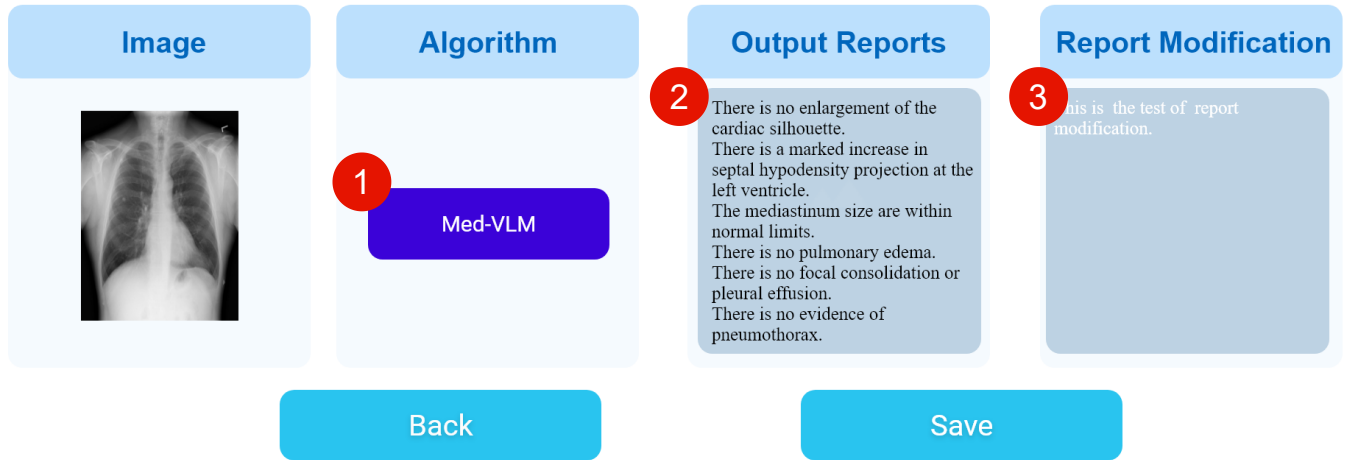


Figure 3: A scenario system for doctors' feedback on report generation quality in the doctor experience assessment. (1) a type of medical vision language model, (2) the output report, (3) the feedback areas from doctors on the generated report

the perceived quality of the report. This score is then used as the reward for the reinforcement learning agent.

the major advantages of the RL-LLM approach is that it eliminates the need for human preference annotations and the training of reward models, which are typically required in traditional reinforcement learning setups. In a conventional reinforcement learning system, human annotators must manually provide feedback on generated reports, and a separate reward model must be trained to predict this feedback. This process is not only time-consuming but also prone to inconsistencies and biases. In contrast, RL-LLM directly uses the LLM's assessment of the quality of the generated content as the reward signal, thus simplifying the learning process and making it more scalable. The large language model, pre-trained on vast amounts of data, is capable of evaluating the quality of generated reports in a way that reflects both linguistic coherence and medical accuracy, ensuring that the reports are of high quality without needing explicit human input at each iteration.

3.5 Model Evaluation

To scientifically evaluate the effectiveness of the adaptive human-LLMs collaborative multimodal vision-language model, a detailed

analysis can be conducted from dimensions such as report quality and doctor experience.

Report quality assessment is a core metric for evaluating the effectiveness of the model. The prevailing methods involve using natural language processing evaluation metrics such as BLEU, ROUGE, and CIDEr, which assess the accuracy, consistency of terminology, and logical coherence of the generated report. Additionally, report completeness assessment is also crucial for checking whether the generated report covers all the key areas and information in the medical image that need to be described.

Doctor experience assessment is an important evaluation metric in the practical use of the model, aiming to determine whether the model can improve the doctor's work efficiency and reduce their workload. The primary methods include assessing interactivity, response speed, and the doctor's satisfaction and acceptance. Interactivity and response speed are used to evaluate the efficiency of the interaction between the doctor and the system, focusing on the system's response time and its real-time feedback capability after doctor input. The doctor's satisfaction and acceptance are collected through surveys or interviews, gathering feedback on the quality of report generation, system usability, and personalized adaptability, evaluating the doctor's acceptance and trust in the model.

Figure 2 illustrates a method for doctors to evaluate the quality of reports generated by a vision-language model. Once the model generates a draft report, doctors review and modify the content to correct any inaccuracies and clarify ambiguous statements. The revised report is then combined with the original medical image that was used to generate the report. This combination of the modified report and the associated medical image forms a feedback sample that serves as a high-quality training example for the system. By incorporating this feedback into the model's training process, the system is able to learn from the doctors' corrections and adjustments, thereby refining its ability to generate more accurate and contextually appropriate reports in the future. The iterative process of doctor-model collaboration enables the model to progressively improve over time by continuously learning from expert input. Each new feedback sample, generated from real-world interactions with medical professionals, allows the model to adjust its understanding of how to interpret medical images and generate corresponding reports that better align with clinical expectations. Figure 3 presents a system implementation of this evaluation method.

4 Conclusion

This paper presents an adaptive Human-LLMs collaborative framework for improving medical report generation by integrating professional doctors' feedback into an AI-driven process. Through the proposed multimodal vision-language model, we address two core challenges in the field: enhancing report quality by incorporating expert feedback and ensuring that the model can dynamically adjust its content without excessive human intervention. The framework incorporates three key modules—the generation module, reinforcement learning module, and large language model distillation module—to achieve a balance between automation and human expertise. By leveraging the strengths of AI in image analysis and the nuanced understanding of medical professionals, our approach aims to significantly improve the accuracy, completeness, and efficiency of medical report generation. The proposed model not only reduces the burden on healthcare providers but also ensures that AI-driven models learn and improve over time, minimizing the need for continuous manual input. This dynamic adaptability holds promise for enhancing the scalability of automated medical report generation models, offering a solution that meets the growing demands of the healthcare industry. Future work will focus on further optimizing these modules and exploring their applicability in real-world clinical settings, ultimately striving to deliver better healthcare outcomes through AI-driven advancements.

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